Exercise Sheet 2

All Lie algebras are real. Write exp for the matrix exponential. Use the matrices

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

For $X \in \mathfrak{sl}_2(\mathbb{R})$ put $\delta = -\det X$. The lecture proved $\exp X = c(\delta)I + s(\delta)X$, with c and s as given there. Exercise 2.7 is an optional extension.

Exercise 2.1 (Rotations in three dimensions). For a unit vector $n = (n_1, n_2, n_3) \in \mathbb{R}^3$, set

$$X_n = \begin{pmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{pmatrix}.$$

(i) Show that $X_n v = n \times v$ for $v \in \mathbb{R}^3$. Deduce $X_n^3 = -X_n$.

(ii) Prove Rodrigues' formula

$$\exp(\theta X_n) = I + \sin\theta\, X_n + (1 - \cos\theta) X_n^2.$$

(iii) Show that $\exp(\theta X_n)$ fixes n, and compute its trace. For $n = (0, 0, 1)$, write the matrix explicitly.

Exercise 2.2 (Boosts and the components of SO(1,1)). Let $\eta = \operatorname{diag}(1, -1)$, let $\mathrm{O}(1,1)$ be as in the lecture, and let $\mathrm{SO}(1,1)$ be its subgroup of determinant one. Write $B(u) = \exp(uK)$.

(i) Prove that every $g \in \mathrm{SO}(1,1)$ equals $\varepsilon B(u)$ for a unique sign $\varepsilon \in \{1, -1\}$ and a unique $u \in \mathbb{R}$.

(ii) Show that $-I \in \mathrm{SO}(1,1)$, and that $-B(u)$ is never an exponential of an element of the Lie algebra $\mathbb{R}K$.

(iii) Explain why (ii) does not contradict generation by exponentials.

Exercise 2.3 (Exponentials in SL(2,R)).

(i) Find all $X \in \mathfrak{sl}_2(\mathbb{R})$ with $\exp X = -I$. Give an example that is not a multiple of J.

(ii) Decide which of the following elements of $\mathrm{SL}(2, \mathbb{R})$ are exponentials of elements of $\mathfrak{sl}_2(\mathbb{R})$:

$$g_1 = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad g_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad g_3 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

For each exponential, give a matrix X. Write g_1 as a product of two exponentials.

(iii) Show that $g_3 = \exp E \exp F$, and that $g_3 \neq \exp(E + F)$.

Exercise 2.4 (SL(2,R) is connected).

(i) Let $g \in \mathrm{SL}(2,\mathbb{R})$. Use Gram–Schmidt on the columns to write $g = kb$ with $k \in \mathrm{SO}(2)$ and $b = \begin{pmatrix} r & x \\ 0 & 1/r \end{pmatrix}$, $r > 0$, $x \in \mathbb{R}$.

(ii) Deduce $g = \exp(\theta J)\exp\big((\log r)h\big)\exp\big((x/r)E\big)$ for some θ.

(iii) Construct a continuous path in $\mathrm{SL}(2,\mathbb{R})$ from I to g, and deduce that $\mathrm{SL}(2,\mathbb{R})$ is connected. You may use that $[0,1]$ is connected.

Exercise 2.5 (The adjoint action of SU(2)). Use the basis of $\mathfrak{su}(2)$ from Lecture 1:

$$A = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix},$$

with $[A,B] = 2C$, $[B,C] = 2A$ and $[C,A] = 2B$. Let $g_\theta = \exp(\theta A)$.

(i) Compute g_θ and the matrix $D(\mathrm{Ad}_{g_\theta})$ in the basis (A,B,C).

(ii) Compute the matrix $D(\operatorname{ad} A)$ of the map $Y \mapsto [A,Y]$ in the same basis. Verify that

$$\exp\big(\theta D(\operatorname{ad} A)\big) = D(\mathrm{Ad}_{g_\theta}).$$

(iii) Find $\theta \in (0, 2\pi)$ with $\mathrm{Ad}_{g_\theta} = \mathrm{id}$. Is $g \mapsto \mathrm{Ad}_g$ injective on $\mathrm{SU}(2)$? Is $Y \mapsto \operatorname{ad} Y$ injective on $\mathfrak{su}(2)$?

Exercise 2.6 (The group commutator). Let $X, Y \in M_{n\times n}(\mathbb{C})$.

(i) Show that

$$\exp(tX)\exp(tY)\exp(-tX)\exp(-tY) = I + t^2[X,Y] + O(t^3).$$

(ii) Use local invertibility in $\mathrm{GL}_n(\mathbb{C})$ to show that the logarithm of this product is $t^2[X,Y] + O(t^3)$. Deduce that a matrix Lie group which is abelian has zero bracket.

(iii) Let G be a connected Lie group whose Lie algebra has zero bracket. Use the cited BCH theorem to prove that G is abelian.

Exercise 2.7 (Optional extension: the third-order term). Let $X, Y \in M_{n\times n}(\mathbb{C})$ and $Z(t) = \log(\exp(tX)\exp(tY))$, so that

$$Z(t) = t(X+Y) + \frac{t^2}{2}[X,Y] + t^3 C + O(t^4).$$

(i) Prove $C = \frac{1}{12}\big([X,[X,Y]] + [Y,[Y,X]]\big)$.

(ii) For $X = E$ and $Y = F$, find $Z(t)$ exactly and compare.